Liquidity, Activism Disclosure, and Takeover Premia

Supplementary Handout

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1 Model Summary

1.1 Notation

Table 1 collects the full notation used in the model. Color coding distinguishes choices made by agents, observables available to the market, and outcomes determined in equilibrium.

Table 1: Model notation. Blue: agent choices. Red: observables. Green: equilibrium outcomes.

Symbol	**Type**	**Description**
Choices / Actions		
$q \in \{-1, 0, +1\}$	Trade quantity	Sell (-1), hold (0), or buy $(+1)$
$a \in \{0, 1\}$	Engagement	Inactive (0) or engage (1)
$z \in \{-1, 0, +1\}$	Noise trade	Exogenous noise trader order
Observables		
$X = q + z$	Order flow	Aggregate order flow observed by market maker
$D = \mathbf{1}\{q = +1\}$	Disclosure	Public filing triggered by net purchase
s	Signal	Blockholder's private signal, $s = v + \varepsilon$
v	Fundamental	Firm value, $v \sim N(\mu, \sigma_v^2)$
Equilibrium Outcomes		
$P(X, D)$	Price	Market-clearing price
$\pi(X, D)$	Posterior	$\mathbb{P}(\text{engaged} \mid X, D)$
$p(P, D)$	Bid probability	Prob. of takeover bid arrival
$m(X, D)$	Premium	Expected takeover premium, $m_0 + (\tilde{m} - m_0)\pi(X, D)$
Parameters		
κ	Liquidity	Noise trading intensity; $z = 0$ with prob $1 - \kappa$
k_1, k_0, k_D	Cutoffs	Exit/Hold, Hold/Quiet, Quiet/Public boundaries
ρ	Success prob.	Prob. engagement succeeds (value improvement and premium wedge realized)
m_0, m_1	Premia	Baseline premium m_0 and premium if engagement succeeds m_1
$\tilde{m}$	Expected premium	Expected premium under engagement, $\tilde{m} = m_0 + \rho(m_1 - m_0)$
$\Delta, \tilde{\Delta}$	Value-add	Standalone improvement if successful; $\tilde{\Delta} = \rho\Delta$
C_0, χ	Cost	Engagement cost scale and signal sensitivity

1.2 Timeline

The model unfolds in four stages:

$t = 0$ **(Nature)** Firm fundamental $v \sim N(\mu, \sigma_v^2)$ is drawn. The blockholder observes a private signal $s = v + \varepsilon$ with $\varepsilon \sim N(0, \sigma_\varepsilon^2)$. The noise trader's order z is drawn from $\{-1, 0, +1\}$ with probabilities $(\kappa/2,\ 1 - \kappa,\ \kappa/2)$.

$t = 1$ **(Trading)** The blockholder chooses (q, a): trade quantity $q \in \{-1, 0, +1\}$ and engagement $a \in \{0, 1\}$. The market maker observes the public signal (X, D) where $X = q + z$ is aggregate order flow and $D = \mathbf{1}\{q = +1\}$ is a disclosure indicator.

$t = 1.5$ **(Bidder Entry)** A potential acquirer draws a private synergy shock ξ and decides whether to bid. The bid probability depends on the price and expected premium: $p(P, D) = 1 - \Phi\big((m(X, D) + K - \bar{S} + P)/\sigma_\xi\big)$.

$t = 2$ **(Payoffs)** If a bid arrives, the takeover occurs at price $P + m(X, D)$. If engagement succeeded ($a = 1$, success probability ρ), the premium is m_1 rather than m_0, and the fundamental improves by Δ.

1.3 Key Assumptions

1. **Discrete trading.** The blockholder's trade $q \in \{-1, 0, +1\}$ is a discrete unit trade, following Back et al. (2018).
2. **Noise structure.** The noise trader's order $z \in \{-1, 0, +1\}$ is symmetric with intensity κ.
3. **Disclosure rule.** Net purchases ($q = +1$) trigger a mandatory public filing ($D = 1$); sales and holds do not.
4. **Engagement cost.** Private engagement costs $C(s) = C_0 \exp(-\chi\,(s - \mu)/\sigma_s)$ decrease with the signal, so better-informed blockholders find it cheaper to engage.
5. **Bidder entry.** The acquirer's entry decision depends on the market price through a feedback channel: higher prices raise the cost of acquisition, reducing bid probability.

1.4 Action Space

The blockholder's four actions are combinations of trading and engagement:

Table 2: Blockholder action space.

Action	**Trade** q	**Engage** a
Exit	-1	0
Hold	0	0
Quiet Voice	0	1
Public Voice	$+1$	1

2 Equilibrium Characterization

2.1 Proposition 1: Monotone Cutoff Equilibrium

Statement. In equilibrium, the blockholder's strategy is characterized by three signal cutoffs $k_1 \le k_0 \le k_D$ such that:

$$(q, a) = \begin{cases} (-1, 0) \quad \text{Exit} & \text{if } s < k_1 \\ (0, 0) \quad \text{Hold} & \text{if } k_1 \le s < k_0 \\ (0, 1) \quad \text{Quiet Voice} & \text{if } k_0 \le s < k_D \\ (+1, 1) \quad \text{Public Voice} & \text{if } s \ge k_D \end{cases}$$

Proof sketch. Single-crossing: each action's expected payoff has a steeper slope in s than the action below it. Exit payoff decreases most slowly in s (sells into a low signal), while Public Voice payoff increases most steeply (buys and engages when the signal is high). Incentive compatibility at each boundary pins down the three cutoffs.

Under the baseline calibration, the Hold region collapses ($k_1 = k_0 \approx 0.82$), so the blockholder transitions directly from Exit to Quiet Voice.

2.2 Proposition 2: Bayesian Posteriors

The market's posterior probability that the blockholder is engaged, $\pi(X, D) = \mathbb{P}(\text{engaged} \mid X, D)$, has two branches:

Disclosure branch ($D = 1$). Disclosure reveals $q = +1$, which implies $a = 1$ (Public Voice). Therefore $\pi(X, 1) = 1$ for all $X \in \{0, 1, 2\}$.

No-disclosure branch ($D = 0$). For $X \in \{-2, -1, 0, 1\}$, define the action probabilities:

$$\omega_E = \Phi\left(\frac{k_1 - \mu}{\sigma_s}\right), \qquad \omega_H = \Phi\left(\frac{k_0 - \mu}{\sigma_s}\right) - \omega_E, \tag{1}$$

$$\omega_Q = \Phi\left(\frac{k_D - \mu}{\sigma_s}\right) - \Phi\left(\frac{k_0 - \mu}{\sigma_s}\right), \qquad \omega_P = 1 - \Phi\left(\frac{k_D - \mu}{\sigma_s}\right), \tag{2}$$

and the noise probabilities $p_0 = 1 - \kappa$, $p_1 = \kappa/2$. Then:

$$\pi(-1, 0) = \frac{\omega_Q \cdot p_1}{(\omega_H + \omega_Q) \cdot p_1 + \omega_E \cdot p_0}, \tag{3}$$

$$\pi(0, 0) = \frac{\omega_Q \cdot p_0}{(\omega_H + \omega_Q) \cdot p_0 + \omega_E \cdot p_1}, \tag{4}$$

$$\pi(1, 0) = \frac{\omega_Q \cdot p_1}{\omega_H \cdot p_1 + \omega_Q \cdot p_1} = \frac{\omega_Q}{\omega_H + \omega_Q}. \tag{5}$$

Note that $\pi(1, 0)$ does *not* depend on κ — it is determined purely by the relative probabilities of Hold and Quiet Voice generating $X = 1$ (both require $z = +1$, so the noise probability cancels).

2.3 Proposition 3: Price Fixed Point

The equilibrium price satisfies the fixed-point equation:

$$P(X, D) = \delta\big[(1 - p^*)\hat{V}(X, D) + p^*(P(X, D) + m(X, D))\big],$$

where $p^* = p(P(X, D), D)$ is the equilibrium bid probability, $\hat{V}(X, D) = \mathbb{E}[v \mid X, D] + \tilde{\Delta} \cdot \pi(X, D)$ is the standalone value under expected engagement improvement, and $m(X, D) = m_0 + (\tilde{m} - m_0)\pi(X, D)$ is the expected premium.

The bid probability depends on the price:

$$p(P, D) = 1 - \Phi\left(\frac{m(X, D) + K - \bar{S} + P}{\sigma_\xi}\right).$$

A contraction mapping argument guarantees existence and uniqueness of the fixed point under regularity conditions on δ.

Economic interpretation. The price reflects a weighted average of the standalone value (if no bid arrives) and the takeover value (if a bid arrives). Because a higher price discourages bidder entry, there is a negative feedback loop: prices partially self-correct, preventing extreme overvaluation.

2.4 Proposition 4: Non-Monotone Liquidity Effect

Statement. The expected minority takeover gains $\Delta^{\min}(\kappa)$ are hump-shaped in liquidity κ.

Two opposing forces:

1. **Bid incidence effect** (rising in κ). Higher liquidity provides more camouflage, enabling Quiet Voice engagement, which raises posteriors π and bid probabilities.
2. **Inference quality effect** (falling in κ). Higher liquidity adds noise, diluting the market's ability to distinguish engaged from non-engaged blockholders, compressing posteriors toward the prior.

At low κ, the first effect dominates; at high κ, the second dominates. The peak occurs at an interior κ^*.

3 Posterior Formulas

Disclosure ($D = 1$). Disclosure requires $q = +1$, which only occurs under Public Voice. The posterior is trivially $\pi(X, 1) = 1$. Order flow $X \in \{0, 1, 2\}$ refines information about z but not about engagement.

No disclosure ($D = 0$). The key inference problem: conditional on $D = 0$, the blockholder chose Exit ($q = -1$), Hold ($q = 0$), or Quiet Voice ($q = 0$, $a = 1$). For each order flow realization X, the market updates using Bayes' rule.

Why $\pi(1, 0)$ is independent of κ. When $X = 1$ and $D = 0$, the blockholder did not buy ($q \neq +1$). So $X = 1$ requires $z = +1$ regardless of the blockholder's action. The only actions compatible with $q \leq 0$ and $X = 1$ are Hold ($q = 0, z = +1$) and Quiet Voice ($q = 0, z = +1$). Both require the same noise realization $z = +1$, so κ cancels from the likelihood ratio:

$$\pi(1, 0) = \frac{\omega_Q \cdot p_1}{\omega_H \cdot p_1 + \omega_Q \cdot p_1} = \frac{\omega_Q}{\omega_H + \omega_Q}.$$

4 Pricing Equation

4.1 Fixed-Point Structure

The price $P(X, D)$ satisfies:

$$P = \delta\Big[\big(1 - p(P, D)\big)\,\hat{V}(X, D) + p(P, D)\,\big(P + m(X, D)\big)\Big]. \tag{6}$$

Components:

- **Standalone value:** $\hat{V}(X, D) = \mathbb{E}[v \mid X, D] + \tilde{\Delta}\,\pi(X, D)$ reflects the fundamental conditional on the information set plus the expected value improvement from engagement.
- **Bid probability:**

$$p(P, D) = 1 - \Phi\left(\frac{m(X, D) + K - \bar{S} + P}{\sigma_\xi}\right). \tag{7}$$

  Since $P(\cdot, D)$ is injective (Assumption A7), the bidder infers X from P; we write $m(X, D)$ as shorthand.

  A higher price raises the effective acquisition cost, reducing the probability that the bidder's synergy shock exceeds the entry threshold.

- **Expected premium:** $m(X, D) = m_0 + (\tilde{m} - m_0)\,\pi(X, D)$. Ranges from m_0 (no engagement) to $\tilde{m}$ (certain engagement, $\pi = 1$).

4.2 Regularity and Contraction

Equation (6) defines P as a fixed point of a map $T(P)$. The derivative $T'(P)$ involves δ and the density $\phi(\cdot)$ evaluated at the bid threshold. A sufficient condition for contraction ($|T'(P)| < 1$) restricts δ to ensure the feedback loop is not explosive. Under the baseline calibration, $T'(P) \approx 0.4$, so convergence is rapid.

5 Numerical Calibration

5.1 Parameter Values

Table 3 reports the baseline calibration.

Table 3: Baseline parameter calibration.

Parameter	**Symbol**	**Value**	**Description**
Discount factor	δ	0.95	Time value of money
Liquidity	κ	0.50	Noise trading intensity
Prior mean	μ	1.00	Mean fundamental value
Fundamental std	σ_v	0.50	Uncertainty in v
Signal noise std	σ_ε	0.50	Noise in private signal
Engagement cost scale	C_0	0.12	Base cost of engagement
Cost sensitivity	χ	0.50	Signal sensitivity of cost
Success probability	ρ	0.90	Prob. engagement succeeds
Engagement value-add	Δ	0.25	Value improvement
Base premium	m_0	0.10	Premium without engagement
Success premium	m_1	0.30	Premium with successful engagement
Baseline synergy	$\bar{S}$	1.44	Mean synergy for bidder
Entry cost	K	0.15	Bidding cost
Synergy shock std	σ_ξ	0.40	Volatility of synergy shock

5.2 Baseline Equilibrium

Under the baseline calibration, the equilibrium cutoffs are $k_1 = k_0 \approx 0.82$ and $k_D \approx 2.26$. The Hold region collapses ($k_1 = k_0$): the blockholder transitions directly from Exit to Quiet Voice. Public Voice is triggered only for very high signals ($s \geq k_D$).

Table 4 reports the full equilibrium price schedule.

Table 4: Baseline equilibrium: prices, posteriors, bid probabilities, and premia across information sets (X, D).

D	Order flow X	$P(X,D)$	$\pi(X,D)$	$p(X,D)$	$m(X,D)$
0	-2	0.84	0.00	0.81	0.10
0	-1	1.06	0.41	0.55	0.17
0	0	1.23	0.74	0.33	0.23
0	1	1.39	1.00	0.17	0.28
1	0	1.90	1.00	0.01	0.28
1	1	1.90	1.00	0.01	0.28
1	2	1.90	1.00	0.01	0.28

6 Figures

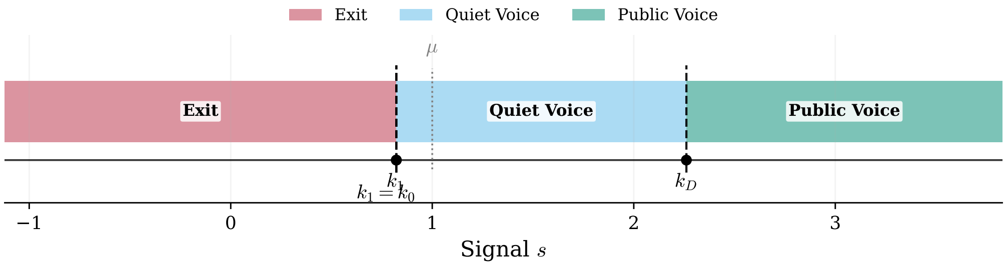

Figure 1: **Equilibrium Cutoff Structure.** Action regions in signal space. The blockholder exits for $s < k_1$, holds for $k_1 \leq s < k_0$ (region collapses under baseline calibration), engages quietly for $k_0 \leq s < k_D$, and engages publicly for $s \geq k_D$. Cutoff locations are determined by incentive compatibility at each boundary.

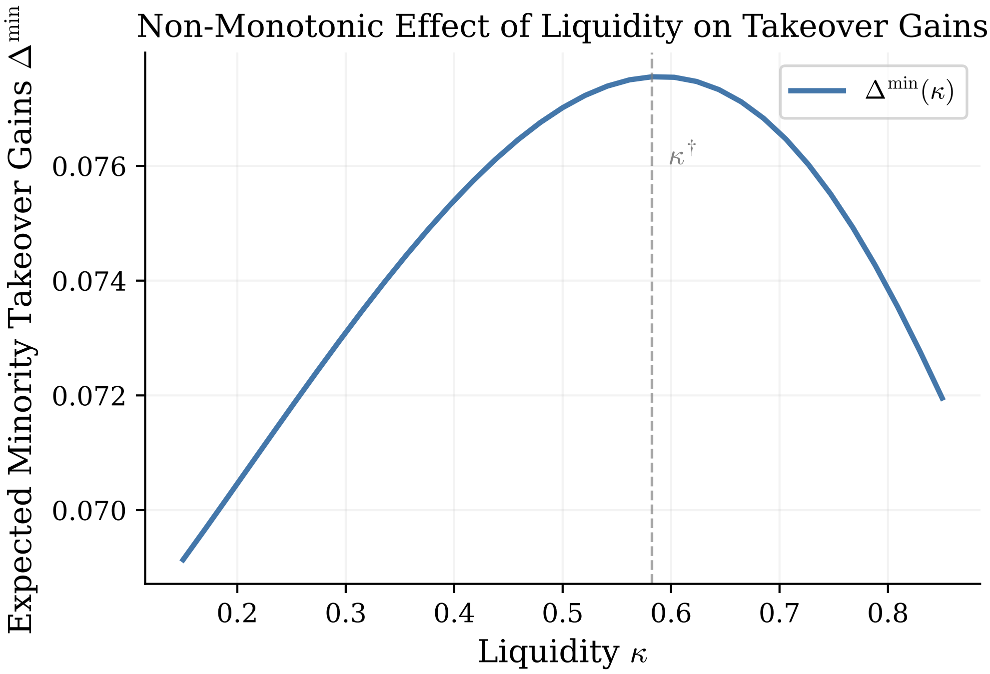

Figure 2: **Non-Monotone Liquidity Effect.** Expected minority takeover gains $\Delta^{\min}$ as a function of liquidity κ. The hump shape reflects two opposing forces: higher liquidity enables camouflaged engagement (raising gains at low κ) but dilutes information quality (reducing gains at high κ). The peak occurs at an interior κ^*.

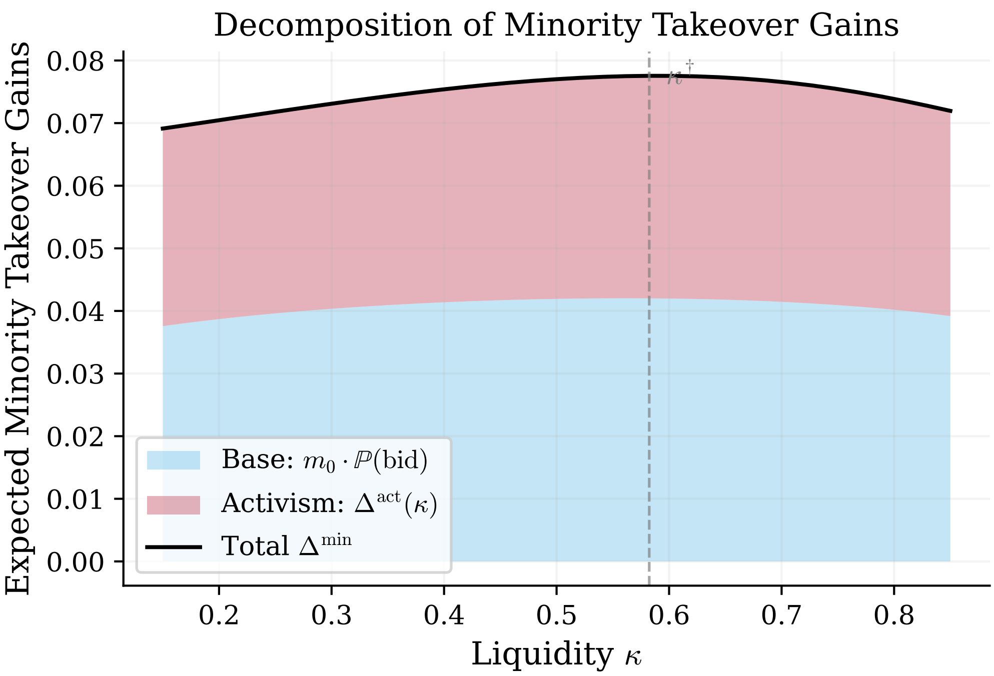

Figure 3: **Decomposition of Minority Gains.** Base premium component $m_0 \cdot \Pr(\text{bid})$ versus activism component Δ^{act}. The base component captures gains available to any minority shareholder; the activism component reflects additional gains from the blockholder's engagement activity and its effect on takeover premia.

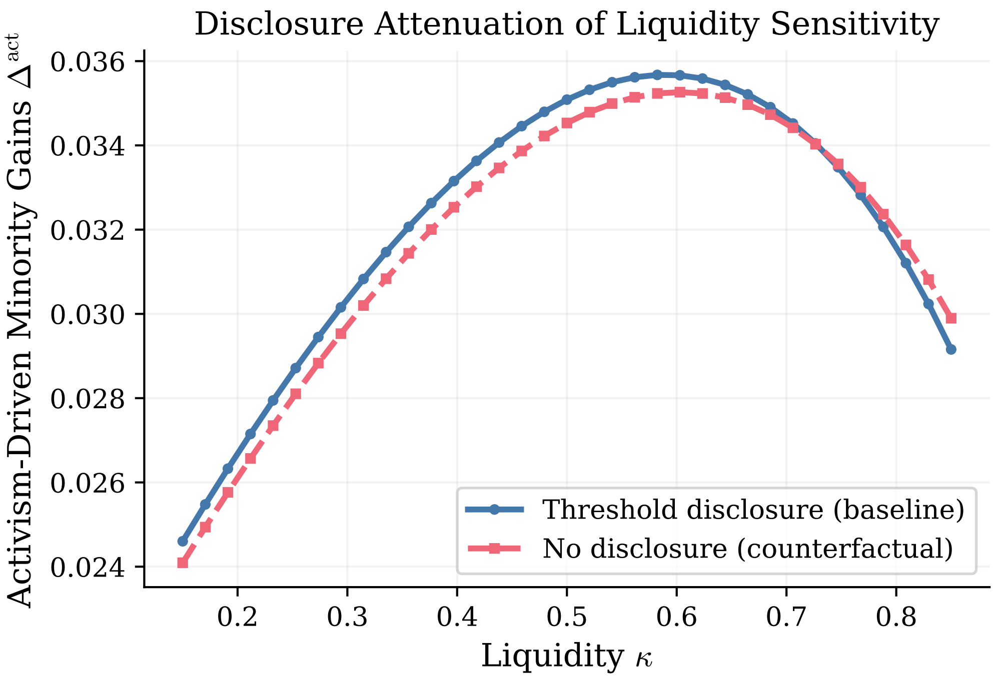

Figure 4: **Disclosure Attenuation.** Comparison of expected minority gains across disclosure states. Disclosure ($D = 1$) fully reveals engagement, eliminating the information asymmetry. The gap between $D = 0$ and $D = 1$ outcomes quantifies the information value of the disclosure channel.

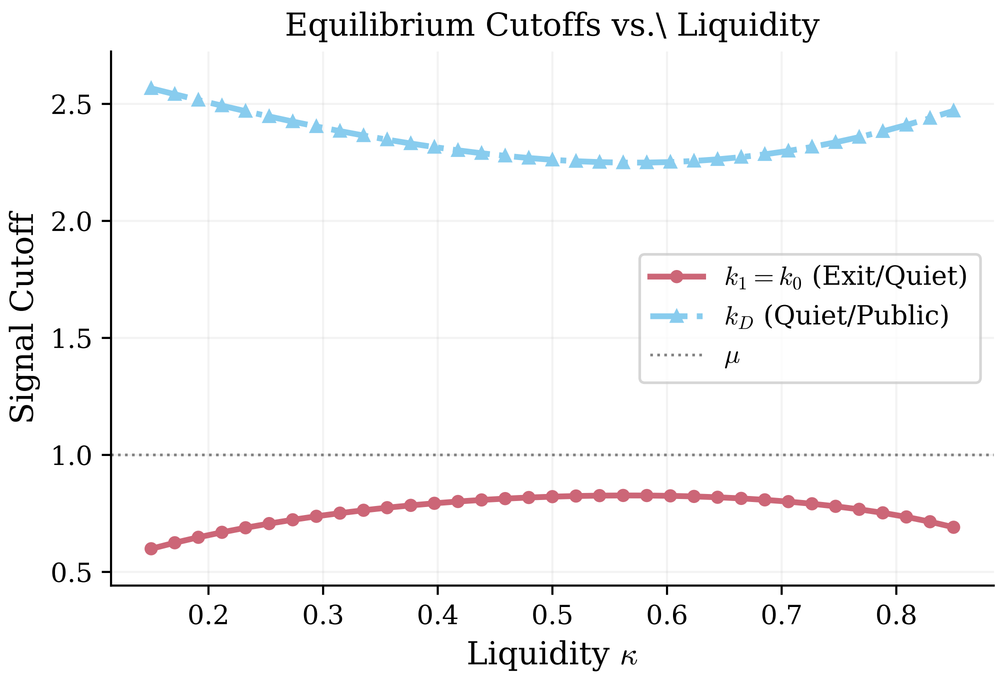

Figure 5: **Cutoffs vs. Liquidity.** How equilibrium boundaries k_1, k_0, and k_D shift with liquidity κ. As κ increases, the noise cover expands the Quiet Voice region and shifts the Public Voice boundary.

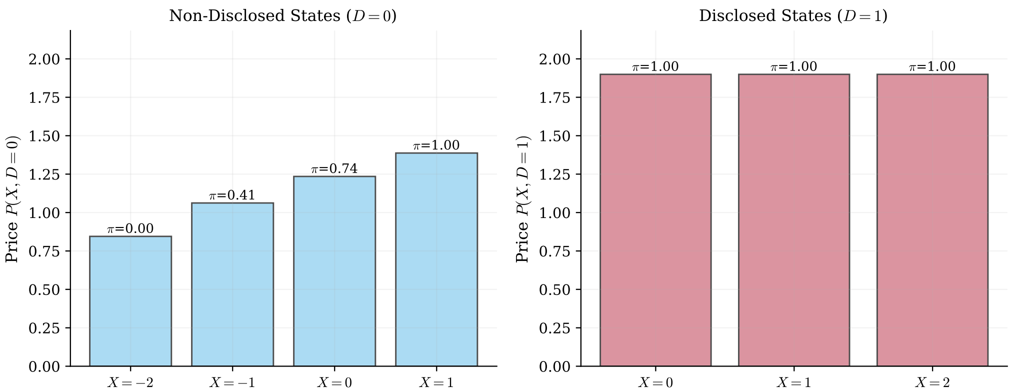

Figure 6: **Price Function.** Equilibrium prices $P(X, D)$ across all information sets. Prices are increasing in order flow X and in the disclosure indicator D, reflecting the market's Bayesian updating about engagement and fundamental value.

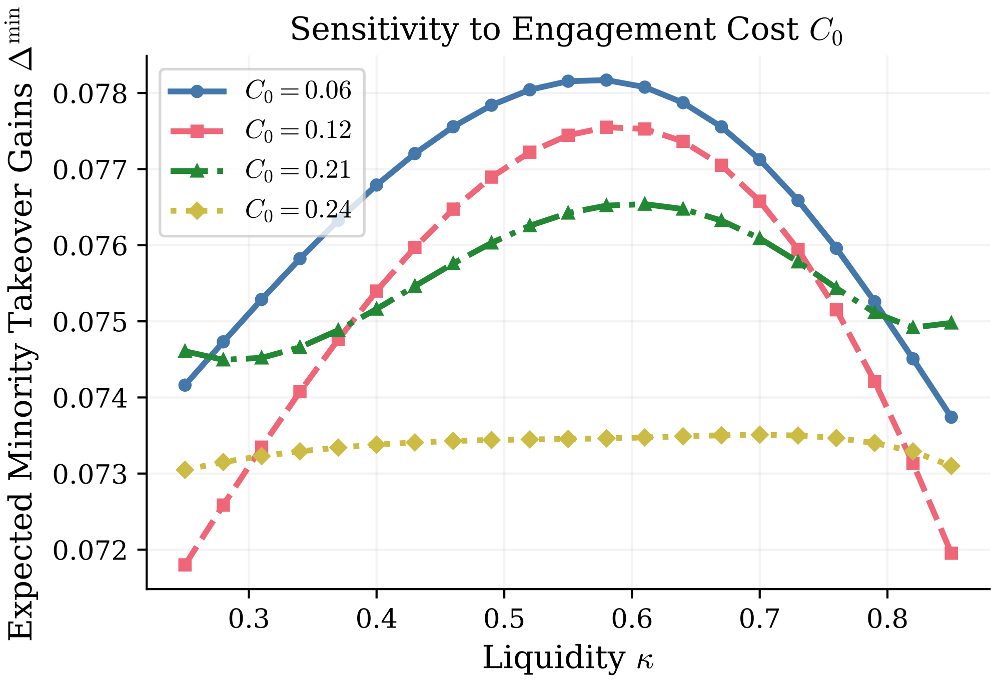

Figure 7: **Sensitivity to Engagement Cost.** Effect of varying C_0 on minority gains. Higher engagement costs shrink the Quiet Voice region, reducing activism-driven gains. Below a threshold C_0, engagement is nearly costless and the blockholder engages for almost all positive signals.

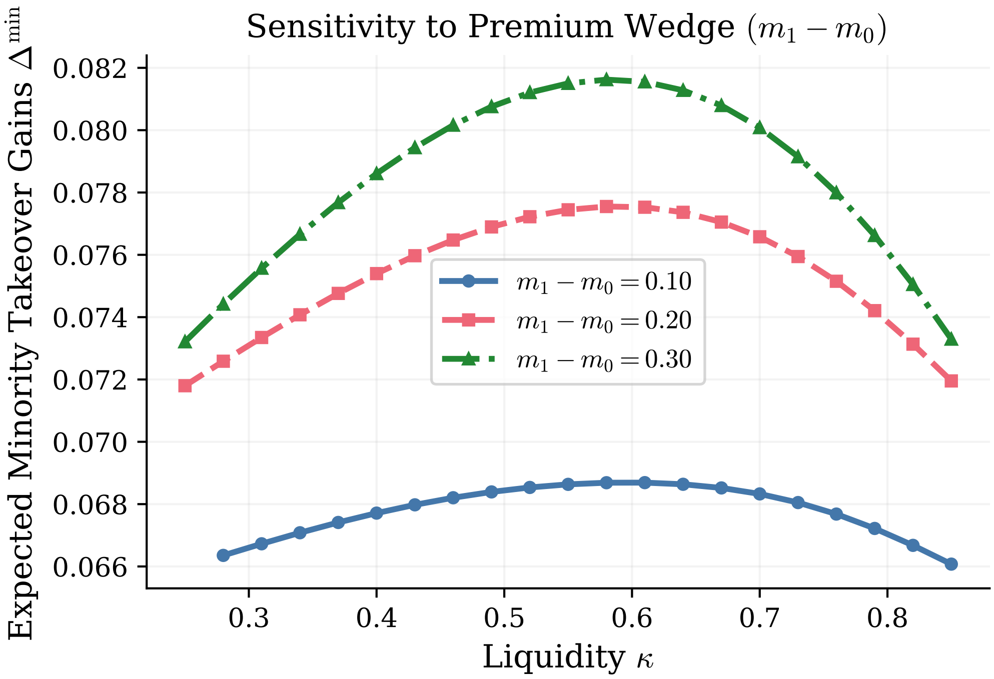

Figure 8: **Sensitivity to Resistance Wedge.** Effect of varying the premium gap $m_1 - m_0$ on minority gains. A larger wedge increases the return to engagement, expanding the Voice region and amplifying the non-monotone liquidity effect. When $m_1 = m_0$, the activism channel shuts down entirely.

7 Extensions: Alternative Disclosure Regimes

The baseline model features a mandatory disclosure rule where net purchases trigger a public filing. Three alternative regimes illustrate how disclosure design interacts with blockholder incentives:

FI (Full Information): The signal $S = (q, a)$ fully reveals the blockholder's action. Inference is shut down: $\pi_{\mathrm{FI}} = 1$ if $a = 1$ and $\pi_{\mathrm{FI}} = 0$ if $a = 0$. This eliminates the strategic interaction between trading and engagement but allows the market to perfectly price the fundamental.

ND (No Disclosure): The signal $S = \varnothing$ reveals nothing beyond order flow. Engagement must be inferred from X alone in all states. Quiet Voice and Public Voice become indistinguishable conditional on the same X, weakening incentives to buy when engaging.

NR (Noisy Rumor): The signal $S = (D, R)$ augments the disclosure indicator with a noisy rumor R that is informative about quiet engagement. This partially bridges the gap between the baseline and full information regimes.

Table 5 compares posteriors across regimes.

Table 5: Posterior comparison across disclosure regimes.

X	$\pi(X, 0)$	$\pi_{\mathrm{ND}}(X)$	$\pi_{\mathrm{NR}}(X, 0, R = 0)$	$\pi_{\mathrm{NR}}(X, 0, R = 1)$
-1	0.41	0.41	0.19	0.68
0	0.74	0.74	0.48	0.89
1	1.00	1.00	1.00	1.00

(q, a)	π_{FI}
$(-1, 0)$	0.00
$(0, 0)$	0.00
$(0, 1)$	1.00
$(1, 1)$	1.00

References

Back, Kerry, Pierre Collin-Dufresne, Vyacheslav Fos, Tao Li, and Alexander Ljungqvist (2018). "Activism, Strategic Trading, and Liquidity". In: *Econometrica* 86.4, pp. 1431–1463.